

A Check-Up for Driving Policies: Counterfactual Exams for End-to-End Planners near Pedestrians

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Abstract—A benchmark score says what a driving policy’s trajectories achieved on logged scenes. It does not say whether a vulnerable road user in the corridor had anything to do with that result. We propose a check-up: a short list of counterfactual exams that read a policy’s behaviour around pedestrians from a few hundred logged frames, without a simulator, a fleet test, or retraining. Each exam edits real frames—removing the pedestrian from the ego corridor, or re-rendering the frame at night—pairs the edit with a control, and has an independent detector verify it. The central reading is not how far the plan moves but how far it moves away: we decompose the response into its magnitude, the policy’s own jitter under an irrelevant edit, and the separation it actually buys from the pedestrian’s logged future. On 246 near-pedestrian scenes across six end-to-end policies, a near-zero response is a reliable certificate that the pedestrian did not enter the decision (it predicts an unchanged outcome in 93–99% of scenes), while a large response is not evidence of avoidance: of the cells in which the plan moved, 43 increased the clearance and 42 decreased it, and only 2.4% of all cells meet all three conditions for genuine avoidance. Across two cities, what transfers is not how strongly a policy reacts but how steadily it drives: the specificity exam predicts the new city’s rate of sub-1.5s time-to-proximity at $\rho = -0.94$, the only significant predictor among the exams and the standard scores alike. We fit how each verdict’s uncertainty shrinks with sample size, so that a check-up can be priced before it is run: the lighting verdict is settled in 14 frames and exposure in 51, while hazard sensitivity needs hundreds. A pre-registered Boston-to-Singapore test confirms the operating range: orderings and verdicts carry over, point values do not.

I. Introduction

End-to-end driving policies are compared by benchmark scores. A score summarises what a policy’s planned trajectories achieved on logged scenes: no collision, staying in the drivable area, making progress [1], [2]. A score of this kind is an outcome measure, and an outcome can be right for the wrong reason. A policy that never approaches a pedestrian because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first time the scene changes. Deciding whether to deploy a policy in a new city requires knowing which of the two it is, and it usually has to be decided from little data: a closed-loop simulator for the target domain, or a fleet test, is exactly what the decision is meant to save.

We propose a check-up: a short list of exams, each with a control and a known normal range, whose results are

read as a diagnosis rather than a ranking (Table I). Every exam is a counterfactual edit of real frames. We remove the pedestrian from the ego corridor, or re-render the frame at night, and read the change in the planned trajectory; an independent detector verifies every edit. The central exam is a hazard-sensitivity curve, measuring how much farther the whole planned trajectory stays from the pedestrian’s logged future because the policy can see the pedestrian, against a model-independent hazard level. A specificity term applies a minimal-change principle: when no reaction is needed, the plan should not move.

a) Displacement is not avoidance: The quantity an edit yields most directly—how far the plan moves when the pedestrian is removed—is a poor safety signal on its own, and treating it as one inverts the ranking. We therefore decompose it into three parts: the magnitude of the response, the displacement the same policy produces under an edit that has nothing to do with the pedestrian (its own jitter floor), and the separation the response actually buys. On 246 near-pedestrian scenes, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes; among the cells where the plan did move, 43 moved away from the pedestrian and 42 moved towards it. The asymmetry that survives is one-directional and useful: a near-zero response certifies that the pedestrian did not enter the decision (93–99% of scenes), while a large response certifies nothing.

b) What the six policies do: Applied to six end-to-end policies, the check-up separates quantities a benchmark score merges (Table II). Their plans cover 0.50–2.19 times the distance the human drove. Seeing the pedestrian improves the plan by 0.01–0.07 on a $[-1, 1]$ scale, and in five of six policies the improvement does not grow with the hazard. Collisions with the pedestrian’s logged future therefore follow how far a policy plans to drive, not what it perceives. A night rendering moves every policy more than the pedestrian does. We look inside the policies to explain this, although we do not attempt a repair: the pedestrian leaves a trace in the features but is not decoded by the policies’ own semantic heads, the planner reads a grid too coarse to locate it, and its plan ends before the pedestrian.

c) What a small check-up can support: A check-up is only useful if it is cheap and if its operating range is known. We fit how each estimate’s spread shrinks with the number of frames: the lighting verdict is settled within 14 frames

and exposure within 51, while hazard sensitivity and the direction of the lighting effect need hundreds. We then pre-registered seven predictions from a diagnosis made on 88 Boston frames and tested them on Singapore. Orderings and verdicts transfer; point values do not (Table VI). Re-measuring the exams in the new city, the exam that predicts behaviour there is not the one measuring how strongly a policy reacts but the one measuring how steadily it drives ($\rho = -0.94$ for specificity).

d) Contributions:

- 1) A small-sample check-up for driving policies: verified counterfactual edits, a hazard-sensitivity curve built on whole-trajectory separation, time to proximity and a minimal-change principle.
- 2) A decomposition of the counterfactual response into magnitude, jitter and separation, which turns a two-sided quantity into a one-sided certificate, together with a diagnosis of six policies and an explanation from inside the networks.
- 3) An account of what small data can and cannot establish: sample-size laws for every exam, a pre-registered cross-city transfer test, and a measurement of which exam predicts behaviour in a new city.

II. Related Work

a) Driving policies and their benchmarks: We examine BEV planners of the TransFuser lineage [3] (LTF, DiffusionDrive [4], DiffusionDriveV2 [5]) and vision-language-action models (SimLingo [6], AutoVLA [7], Alpamayo-1.5 [8], [9]). NAVSIM [1], [2] scores their plans by multiplying hard gates with a weighted average of soft terms. Open-loop scores are known to reward proxies [10]. We show that the six-policy spread is carried by a single gate.

b) Behavioural testing: Checklists of targeted behavioural tests [11] and corruption benchmarks [12]–[14] probe models beyond aggregate accuracy. A metric drop, however, does not say whether a policy became more cautious or more reckless. Our exams read signed plan changes against a counterfactual and state how many frames each needs.

c) Causal confusion: Imitation learners can respond to correlates of the expert’s action rather than to its causes [15]–[18]. Our exams apply this idea to trained policies: a cause (a pedestrian) and a nuisance (lighting) are intervened on directly.

d) Counterfactual editing and internal representations: Counterfactual explanations edit the scene [19], [20]; we verify every edit with a detector. Steering directions [21] can show that a network encodes what it does not act on [22]; we use them to explain a diagnosis, not as the diagnosis.

TABLE I

The check-up. Each exam, the question an investor would ask, the counterfactual that answers it, and the normal range of a driver that yields.

Exam	Investor’s question	Measurement	Normal
Exposure	Position size	planned / logged distance, 2.5 s	≈ 1
Hazard sensitivity HS	Risk management	safety gained by seeing the pedestrian, when needed	rises to 1 with hazard
Scaling	Procyclicality	rank corr. of HS with hazard	> 0
Specificity SP	Over-reaction	plan change when no reaction is needed	≈ 1
Lighting CFR, align	Reaction to noise	night vs. removal, same frame	CFR $\gg 1$, align ≈ 0

III. The Check-Up

A. Counterfactual units

A unit is a logged frame together with an input ego speed. We use the logged speed for every frame, and for frames from approach sequences also counterfactual speeds of 2–8 m/s. For each unit we query the policy three times. The query uses its full input (all cameras, history frames, LiDAR, ego status), edited in one of three ways: unchanged (P), with the target pedestrian removed (Q), or re-rendered at night (N). All policies are read on their common horizon of 2.5 s. Plans and the pedestrian’s logged future positions are interpolated to 0.1 s.

a) Removal: We project the pedestrian’s annotated 3D box into the camera; dataset 2D boxes proved mis-associated in an earlier corpus. We then segment the pedestrian with SAM [23] (box and torso prompts, clipped to the box, dilated by 9 px) and fill the mask with LaMa [24]. For the LiDAR policy we also delete the points inside the box. The VLA models see several cameras, of which we edit the front one, so we keep only frames in which the target is visible in no other camera.

b) Night: A closed-form photometric transform (gamma 2.2, gain 0.45, saturation 0.45, blue tint, sensor noise $\sigma = 7$) is applied to every image the policy consumes. The noise seed is fixed per frame.

c) Validity: An independent detector (Faster R-CNN [25], COCO, score ≥ 0.5) finds the target in the original image of 248 front-only frames. After removal it still finds the target in 8.5% of them ($\text{IoU} \geq 0.5$); these failures are residual legs, bags, or companions who overlap the target. After the night rendering it finds the target in 89.9%. The inpainting artefact, calibrated by pasting patches on empty road, is 5–17% of the removal effect at pedestrian scale. Removing off-road pedestrians, which should not matter causally, serves as a same-type control.

B. Readouts: the whole trajectory against the pedestrian

Let $c_t(X) = \|X_t - F_t\| - 1.0 - 0.4$ be the clearance at time t between plan X and the pedestrian’s logged position F_t ,

net of the ego half-width and a pedestrian radius. We read four properties of a plan:

- its separation $S(X) = \text{mean}_t \min(c_t, 10 \text{ m})$, i.e., how far the whole trajectory stays from the pedestrian;
- contact $A(X) = \mathbb{1}[\min_t c_t > 0]$;
- safety distance $C(X) = \text{clip}(\min_t c_t / 1 \text{ m}, 0, 1)$;
- time to proximity $T(X)$, the first time $c_t < 1 \text{ m}$, divided by 2.5 s (1 if it never happens).

Speed changes are deliberately not a readout. The safety of a plan is judged by where it goes relative to the pedestrian, not by how it decelerates.

C. Exams

a) Exposure: The ratio of planned to logged distance over 2.5 s, on frames where the human moved at least 0.5 m.

b) Hazard sensitivity: A unit needs a reaction if the pedestrian-blind plan comes within 1 m of the pedestrian, $C(Q) < 1$. On such units,

$$\text{HS} = \frac{1}{4} [\Delta A + \Delta C + \Delta T + \tanh(\Delta S / 1 \text{ m})] \in [-1, 1], \quad (1)$$

with $\Delta = (\cdot)(P) - (\cdot)(Q)$: the safety that seeing the pedestrian buys. We plot HS against the model-independent hazard level $a_{\text{req}} = v^2 / 2 \max(d - 2, 0.5)$, the deceleration needed to stop before a pedestrian at distance d , and alternatively against $\text{TTC}_0 = (d - 2) / v$. Units with input speed below 1 m/s are excluded, because a stationary ego cannot move toward or away from anyone.

c) Scaling and specificity: Scaling is the rank correlation of HS with a_{req} across needed units. Specificity applies the minimal-change principle to the units that need no reaction: $\text{SP} = \exp(-\bar{D}(P, Q) / 0.5 \text{ m})$, where $\bar{D}$ is the mean displacement between the two plans.

d) Lighting: For the same frames, $\text{CFR} = \mathbb{E} \|P - Q\| / \mathbb{E} \|P - N\|$ compares the plan change caused by removing the pedestrian with that caused by the night rendering. Alignment is the median cosine between $N - P$ and $Q - P$. A pure policy has $\text{CFR} \gg 1$ and alignment near zero. Positive alignment means the night pushes the plan the way removing the pedestrian does.

D. Policies and frames

We examine LTF and DiffusionDrive (front camera and ego status), DiffusionDriveV2 (camera and LiDAR), SimLingo (InternVL2-1B), AutoVLA (Qwen2.5-VL-3B; three cameras $\times$ four frames) and Alpamayo-1.5 (four cameras; a diffusion head over the VLM cache). All use released weights. The frames are nuScenes [26] keyframes. They cover corridor pedestrians (lateral clearance to the ego corridor $\leq 1 \text{ m}$) at 5–12 m, their off-road counterparts, and 30 approach sequences at 5.8–14.6 m with logged pedestrian futures. The near-pedestrian units come from 236 front-only frames, 88 in Boston (left-hand drive) and 148 in Singapore (right-hand drive). Alpamayo-1.5 covers 170 of the 236, since it needs 1.6 s of history and 6.4 s of future within the scene. We fix DiffusionDrive’s diffusion seed so that paired queries share their noise.

IV. Diagnosis of Six Policies

A. The report

Table II and Fig. 1c give the six profiles.

a) How far they plan to drive: Exposure spans a factor of four. DiffusionDrive plans half the human’s distance; SimLingo plans more than twice it; AutoVLA and Alpamayo-1.5 plan about as far as the human.

b) Whether they yield: No policy manages the pedestrian risk. On units that need a reaction, seeing the pedestrian buys 0.01–0.07 of safety on a $[-1, 1]$ scale. The benefit does not grow with the hazard: its rank correlation with a_{req} is negative for five policies, down to -0.51 for Alpamayo-1.5. DiffusionDriveV2 is the only exception ($+0.29$), and its benefit stays below 0.15. The time-based hazard axis gives the same picture (Fig. 2c). Counting only cases that change outcome, seeing the pedestrian turns a collision into a clean pass about as often as it does the reverse:

- DiffusionDrive: 9 passes vs. 5 collisions;
- SimLingo: 18 vs. 30;
- the other four policies: at most 5 either way;
- all McNemar $p \geq 0.11$.

c) Over-reaction: SimLingo’s plan moves most when nothing requires it ($\text{SP} = 0.42$). The other policies stay at 0.67–0.79.

d) Outcome: Collisions with the pedestrian’s logged future follow how far a policy plans to drive, not what it perceives. At the logged speed, the visible and removed rates differ by at most 0.4 points for every policy, and at 8 m/s by at most 3.8. SimLingo, which plans the longest trajectories, collides most. The low-exposure policies almost never reach the pedestrian. DiffusionDrive is safe because it stays short, AutoVLA and Alpamayo-1.5 cover a human-like distance without yielding, and SimLingo covers the most ground with the least stable plan.

B. Looking inside: why the risk goes unmanaged

The check-up only diagnoses; the following explains the diagnosis for the three BEV planners (Table III, Fig. 2).

a) The pedestrian leaves a trace but is not decoded: The planners’ own BEV semantic heads do not localise the pedestrian. On 520 NAVSIM near-pedestrian scenes, the pedestrian channel is no brighter at the pedestrian than at a mirrored empty location (Table III, top). It is 20–55 times weaker than the vehicle channel and never wins the argmax inside a pedestrian’s box. The pedestrian does leave a trace in the features, however. The direction between SimLingo’s representations with and without the pedestrian is steerable: it shortens the plan by 2.9 m at the largest dose, while random directions lengthen it. Location-specific information also survives in the 1 m side branch, as shown next.

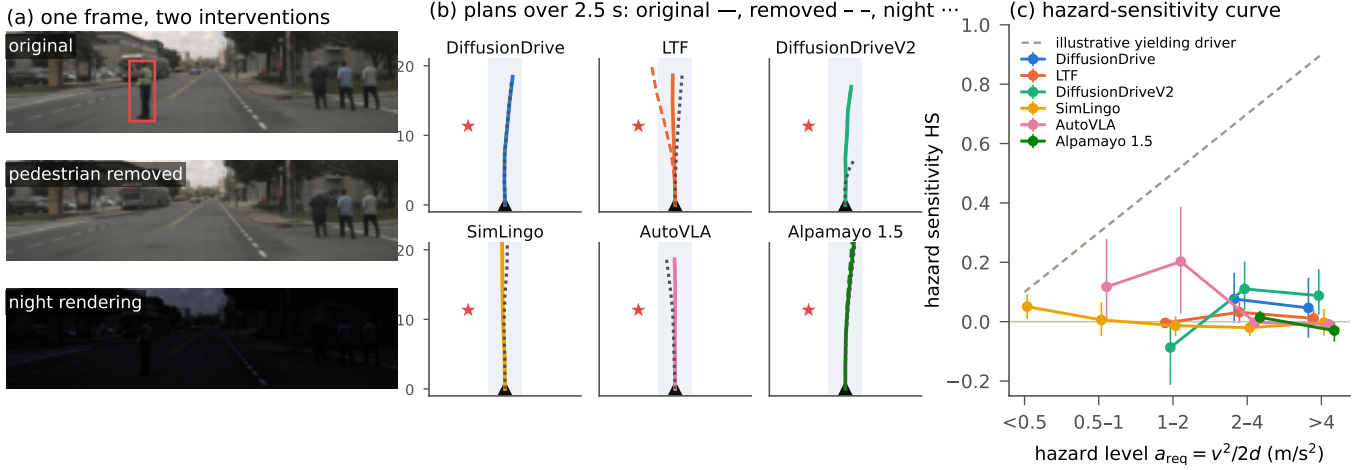


Fig. 1. One frame, two edits, six policies, and the resulting curve. (a) A worker in a high-visibility vest stands at the edge of the ego corridor 9.6m ahead; the ego vehicle moves at 7.6m/s. The detector finds the worker in the night image but not in the removed one. (b) Plans over 2.5s: original (solid), pedestrian removed (dashed), night (dotted); the star marks the worker. (c) Hazard-sensitivity curves over all near-pedestrian units that need a reaction (mean, 95% CI, bins with ≥ 8 units). A policy that yields would rise toward 1 as hazard grows; all six stay near zero.

TABLE II

The check-up report of six policies (pooled near-pedestrian units, both driving sides; 95% bootstrap intervals over frames). Exposure: planned / logged distance over 2.5s (1 = human). HS: safety gained by seeing the pedestrian when it matters, on $[-1, 1]$ (a yielding driver approaches 1). Scaling: rank correlation of HS with hazard level a_{req} (does it respond more when risk is higher). SP: how little the plan moves when no reaction is needed (over-reaction; 1 = unmoved). Collision: contact with the pedestrian’s logged future, pedestrian visible / removed, at the logged speed and at 8 m/s. Right: the NAVSIM benchmark on 783 Singapore scenes.

Policy	exposure	HS	scaling	SP	collision (%), visible / removed		NAVSIM	
	(distance)	(yielding)	(scaling)	(over-reaction)	logged v	8 m/s	EPDMS	DAC
DiffusionDrive	0.50 [0.46,0.55]	+0.07 [-0.01,0.14]	-0.25	0.78 [0.75,0.81]	0.0 / 0.0	7.6 / 9.2	0.535	0.673
LTF	0.79 [0.75,0.84]	+0.02 [-0.01,0.04]	-0.03	0.79 [0.76,0.81]	0.4 / 0.4	9.9 / 10.7	0.509	0.630
DiffusionDriveV2	0.88 [0.77,0.95]	+0.06 [0.01,0.11]	+0.29	0.67 [0.63,0.71]	0.4 / 0.0	3.1 / 6.1	0.652	0.820
SimLingo	2.19 [1.98,2.65]	+0.01 [-0.02,0.03]	-0.19	0.42 [0.36,0.48]	17.4 / 17.4	40.5 / 36.6	0.359	0.452
AutoVLA	1.01 [0.97,1.11]	+0.02 [-0.00,0.05]	-0.18	0.76 [0.72,0.79]	1.7 / 1.3	32.1 / 29.8	0.683	0.814
Alpamayo 1.5	1.13 [1.06,1.21]	+0.01 [-0.01,0.02]	-0.23	0.84 [0.81,0.87]	0.0 / 0.0	11.2 / 11.2	0.663	0.789

b) The planner cannot locate it: The trajectory head reads an 8×8 grid at 8m per cell. In 73% of 94 pairs of corridor and off-road pedestrians, the two fall in the same or an adjacent cell. A corridor-minus-off-road direction injected there does not beat random directions ($p = 0.27$). At the 1m side branch it does (52-fold, $p = 0.007$), but with $\sim 1/460$ of the effect on the trajectory; in LTF that branch does not reach the trajectory at all.

c) The plan does not reach the pedestrian: Plans end 5–9m ahead, while the median corridor pedestrian stands 20m away; only 5–11% of pedestrians are within reach. When a distance sweep over all 850 nuScenes scenes brings the pedestrian within 12m, LTF begins to respond (Fig. 2a). At 8–12m its plan changes by 0.23m ($p = 5 \times 10^{-7}$), which separates corridor from off-road pedestrians after mask area is regressed out (AUC 0.65; 0.67 on an independent draw). Raising the input speed removes that response again, and collisions then rise identically with and without the pedestrian in view (Fig. 2b). DiffusionDrive

never responds.

C. What the benchmark sees

On 783 NAVSIM scenes scored identically, 95% of the between-policy variance of EPDMS comes from the two map-compliance gates (drivable area 76%, driving direction 19%); the no-collision gate contributes 1% (Table II). The benchmark ranks these policies by map compliance, and none of the exams above is visible in it.

V. Against the Standard Evaluations

How does the check-up relate to the evaluations these policies are usually judged by? We ran both on the same six policies (Table IV).

a) nuScenes open-loop metrics: On our 236 near-pedestrian frames, removing the pedestrian changes the L2 error by at most 0.07m, frontal collisions by at most 1.7 points and pedestrian collisions by at most 3.0. The metric cannot tell whether a policy saw the pedestrian. Its ordering follows how far a policy’s planned distance

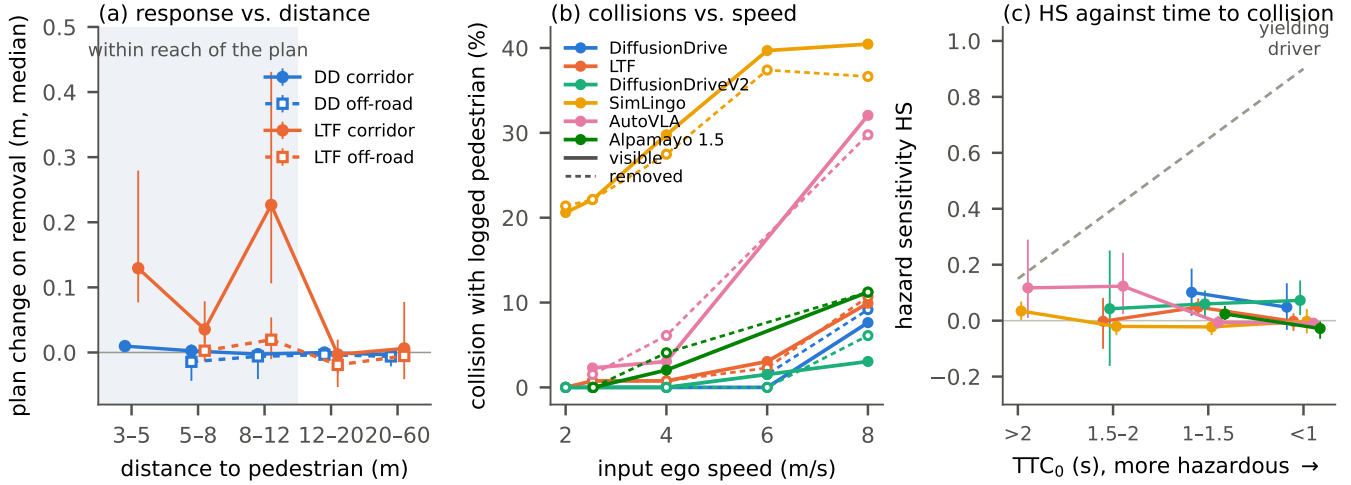


Fig. 2. Why the risk goes unmanaged, and what it costs. (a) Plan change when the pedestrian is removed, by distance, across all 850 nuScenes scenes (974 pedestrian records; median, 95% CI). LTF responds to corridor pedestrians (filled) once they are within reach of its plan; off-road pedestrians (open) evoke nothing; DiffusionDrive is flat throughout. (b) Collision with the pedestrian’s logged future along the plan, at the logged speed and at counterfactual input speeds; solid: pedestrian visible, dashed: removed. Each pair of curves nearly coincides, and every curve rises with speed. (c) Hazard sensitivity against the time-based hazard axis TTC_0 (bins with ≥ 8 units); colours as in (b).

TABLE III

Not decoded, not transmitted, and out of reach (BEV planners).

Top: the policy’s own BEV semantic head on 520 NAVSIM near-pedestrian scenes (in parentheses: the same readout at a mirrored empty location). Middle: difference-of-differences injection (corridor minus off-road pedestrian direction), reported as effect over the median of random directions, with permutation p . Bottom: planning reach.

	DD	LTF	DDv2
Ped. peak at pedestrian / global	0.07 (0.11)	0.07 (0.07)	0.09 (0.15)
Ped.-box px won by ped.	0%	0%	0%
Pedestrian / vehicle peak	0.018	0.023	0.052
Injection, 8 m grid (plan-ner)	$2.9 \times (p=.27)$	$p=1.0$	–
Injection, 1 m grid (branch)	$52 \times (p=.007) \equiv 0$	$14/14$	–
effect vs. 8 m grid	$\sim 1/460$	–	–
Median plan length (m)	5.3	7.1	8.9
Median pedestrian distance (m)	20.8	20.8	20.6
Pedestrian within plan reach	5%	11%	10%

departs from the human’s (rank correlation $+0.66$ with $|\text{exposure} - 1|$): L2 rewards driving at the human’s speed. We exclude rear-end contacts from the collision count. The logged traffic does not react, so a policy that plans to drive slower than the human is scored as being hit from behind.

b) NAVSIM, sliced to near pedestrians: On all 783 Singapore scenes, AutoVLA, Alpaymayo-1.5 and DiffusionDriveV2 score highest. On the 260 scenes with a pedestrian near the ego corridor, the ordering is unrelated to that

one ($\rho = -0.09$): the leaderboard’s first (AutoVLA) falls to fifth and its third (DiffusionDriveV2) to last, while its fourth and fifth (DiffusionDrive, LTF) rise to second and third. In these scenes the no-collision gate carries 100% of the spread, against 1% over all scenes. The cause is a trait the check-up had not examined: 63% of these scenes show a human driver waiting. DiffusionDriveV2 and AutoVLA plan to move off and run into the stopped traffic. On our own frames where the human waits, DiffusionDriveV2 plans to move off in 81% of cases, against 3% for DiffusionDrive and 0% for Alpaymayo-1.5. This agrees with NAVSIM for the extremes but not for AutoVLA. Of our exams, specificity predicts the near-pedestrian ordering ($\rho = +0.66$), while hazard sensitivity does not ($+0.14$), because no policy has any.

c) Pedestrian TTC across driving sides: Following the plan against the pedestrian’s logged future, we count the moving-ego scenes in which the minimum time to collision falls below 1.5 s. Near-pedestrian scenes from the left-hand-drive cities (nuScenes Boston and NAVSIM Las Vegas, Boston, Pittsburgh) and from Singapore give orderings that agree at $\rho = +0.77$. Both orderings coincide with planned distance ($\rho = +1.00$ and $+0.77$ with exposure). They agree only partly with hazard sensitivity ($+0.83$, $+0.49$), which is near zero for every policy, and not with lighting invariance ($+0.14$, -0.43). NAVSIM’s own TTC term, which counts all objects over 4 s, agrees only weakly (0.31 – 0.54).

d) Which evaluation predicts behaviour in another domain?: This is the question a deployment decision has to answer, so we pose it directly in Table V. The target is the ordering of the six policies on the Singapore near-

TABLE IV

The same six policies under the standard evaluations. Left: nuScenes open-loop metrics on the same 236 near-pedestrian frames, original / pedestrian removed (L2 to the human trajectory averaged over 0.5–2.5 s; collisions with objects ahead of the ego, rear-end contacts by non-reactive logged agents excluded). Middle: NAVSIM EPDMS on all 783 Singapore scenes and on 260 Singapore scenes with a pedestrian within 1 m of the ego corridor and 20 m ahead. Right: pedestrian TTC along the plan (share of moving-ego scenes with minimum TTC < 1.5 s, lower is better) on left-hand-drive near-pedestrian scenes (nuScenes Boston + NAVSIM Las Vegas/Boston/Pittsburgh) and right-hand-drive ones (NAVSIM Singapore), with the resulting rank.

Policy	nuScenes open-loop (orig. / removed)			NAVSIM EPDMS		pedestrian TTC < 1.5 s (rank)	
	L2 (m)	collision (%)	ped. coll. (%)	all	near ped.	left-hand	right-hand
DiffusionDrive	0.98 / 0.99	14.0 / 14.8	1.3 / 1.7	0.535	0.744	3.2 (1)	14.1 (1)
LTF	0.88 / 0.91	4.2 / 3.8	2.5 / 1.7	0.509	0.735	5.4 (2)	16.5 (2)
DiffusionDriveV2	1.35 / 1.31	6.4 / 7.2	3.8 / 3.8	0.652	0.367	18.3 (3)	22.4 (5)
SimLingo	3.28 / 3.35	37.3 / 35.6	22.5 / 19.5	0.359	0.583	61.3 (6)	56.5 (6)
AutoVLA	0.68 / 0.66	6.4 / 5.9	3.8 / 3.4	0.683	0.413	19.4 (4)	21.2 (4)
Alpamayo 1.5	0.77 / 0.72	2.4 / 2.4	0.6 / 0.6	0.663	0.760	25.4 (5)	20.0 (3)

TABLE V

Which evaluation predicts behaviour near pedestrians in another domain? Rank correlation between the six-policy ordering given by each evaluation (none uses the target data) and the ordering observed on 260 NAVSIM Singapore scenes with a pedestrian near the ego corridor (original images, moving ego). Bold: $|\rho| \geq 0.89$ (two-sided 5% for $n=6$).

Ordering from	Singapore near-pedestrian ordering by			
	TTC	clearance	collisions	EPDMS
Same readouts, nuScenes Boston				
TTC violations	+0.94	+0.83	+0.94	+0.14
clearance	+0.83	+0.94	+0.83	+0.60
collisions	+0.83	+0.94	+1.00	+0.09
Same readouts, NAVSIM US cities				
TTC violations	+0.83	+0.94	+1.00	+0.09
clearance	+0.83	+0.71	+0.89	−0.03
near-ped. EPDMS	+0.77	+0.54	+0.43	+0.89
nuScenes open-loop (Boston)				
L2	+0.43	+0.09	+0.03	+0.49
collision	+0.26	−0.09	−0.03	+0.03
ped. collision	+0.60	+0.37	+0.26	+0.71
NAVSIM leaderboard				
EPDMS, 783 scenes	+0.14	−0.09	−0.03	−0.09
Check-up (Boston)				
exposure	+0.83	+0.94	+1.00	+0.09
hazard sensitivity	−0.66	−0.89	−0.94	+0.09
specificity	+0.83	+0.71	+0.60	+0.77
lighting CFR	−0.20	−0.09	+0.14	−0.94
patience	+0.31	−0.03	−0.14	+0.77

pedestrian scenes (original images), by TTC violations, clearance to the pedestrian, pedestrian collisions and EPDMS. Each predictor is an ordering obtained without that data. The standard evaluations do not predict it: nuScenes L2 and collision rates, and the NAVSIM leaderboard score, correlate with it at −0.54 to +0.49. The same pedestrian-centred readouts taken on Boston or on the US cities do predict it (0.83–1.00), and so does the check-up’s exposure exam (0.94). Near-pedestrian EPDMS is predicted best by near-pedestrian EPDMS elsewhere (0.89) and by specificity (0.77). Hazard sensitivity and lighting invariance measure something else, and they do not predict these orderings.

e) Reading: Where a standard metric measures what an exam measures, they agree (pedestrian collisions: $\rho = +0.94$; TTC and exposure). None shows whether the pedestrian changed the plan, and the full-set score hides the near-pedestrian scenes.

VI. How Much Data Does a Check-Up Need?

A. Fitting

For every exam and policy we subsample n of the 170 frames that all six policies share, 400 times, and fit the spread of the estimate with $\sigma(n) = a\sqrt{1/n - 1/N}$ (Fig. 3).

a) Well-fitting exams: Exposure, the lighting ratio and specificity obey this law ($R^2 = 0.97$ –1.00). The fitted a converts directly into the number of frames a verdict needs.

- Lighting outweighs the pedestrian (CFR < 1) is settled for every policy within 14 frames.
- Exposure differs from the human’s is settled within 51 frames for the four policies whose exposure is far from one. For AutoVLA and Alpamayo-1.5 it would take hundreds to thousands of frames; this is itself a finding, since they plan a human-like distance.
- Orderings of the six policies stabilise quickly: exposure with 10 frames (96% of subsamples reach $\rho \geq 0.6$ with the full ordering) and specificity with 40 (99%).

b) Poorly fitting exams: Hazard sensitivity and the direction of the lighting effect behave differently. Their fits are poor (R^2 as low as 0.26 and −0.09) wherever few units need a reaction. Deciding that HS $\neq 0$ needs 57–1,072 frames, and the ordering reaches 84% stability only at 90 frames. The ordering by alignment is unstable even at 130 (72%).

The division is therefore clear. A few dozen frames diagnose planned distance, over-reaction and sensitivity to lighting. Whether a policy manages hazard, and in which direction lighting pushes it, needs hundreds of frames that contain a genuine need to react.

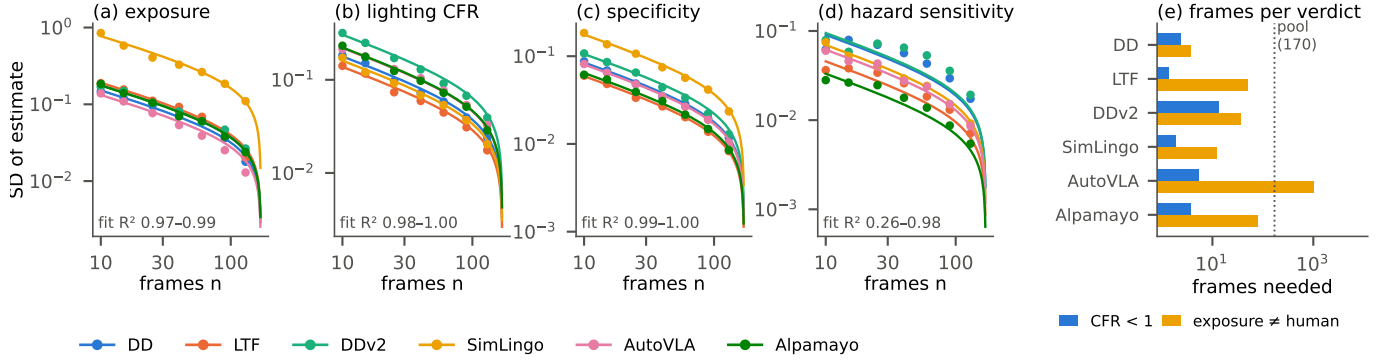


Fig. 3. Fitting estimator spread against sample size. (a–d) Points: standard deviation of each exam’s estimate over 400 subsamples of n frames, drawn without replacement from the 170 frames all six policies share. Lines: the fitted finite-population law $a\sqrt{1/n - 1/N}$. Exposure, lighting and specificity follow it closely. Hazard sensitivity fits poorly for policies with few units that need a reaction. (e) Frames needed to place the estimate two standard deviations from the verdict threshold, extrapolated from the fit. For AutoVLA and Alpamayo-1.5, exposure is indistinguishable from the human’s.

TABLE VI

Pre-registered transfer test. Diagnoses made on 88 left-hand-drive frames (Boston) predict the near-pedestrian units of 148 right-hand-drive frames (Singapore). Predictions and decision rules were fixed before any right-hand statistic was computed. Evaluated on the five policies whose registration the Alpamayo-1.5 replacement leaves intact (Section VI); retaining the replaced policy changes no verdict.

Prediction	Holds	Evidence
P1 Lighting outweighs the pedestrian (CFR < 1) for every policy	yes	max upper CI 0.70
P3 Exposure ordering transfers ($\rho \geq 0.6$; SimLingo top, DD bottom)	yes	$\rho = 0.90$
P4 Over-reaction ordering transfers ($\rho \geq 0.6$; SimLingo lowest SP)	yes	$\rho = 0.90$
P7 SimLingo collides most; visible vs. removed within 3 pts	yes	gap 2.7 pts
P2 Hazard sensitivity stays below 0.15 (upper CI)	no	fails for DD, DDv2
P5 Point values stay within the left-hand CI ($\geq 70\%$ of 15 cells)	no	60%
P6 Purer policies (lower align) shift less	no	$\rho = -0.90$ (opposite)

B. A pre-registered transfer test

Does a diagnosis made in one city describe the same policy in another? Before computing any right-hand-drive statistic, we fixed seven predictions and their decision rules. They were derived from the profile of the 88 Boston frames, and we tested them on the near-pedestrian units of the 148 Singapore frames (Table VI). The registration covered the six policies public at that time. NVIDIA subsequently released Alpamayo 1.5, which accepts navigation instructions, and we use it in place of Alpamayo-R1 throughout this paper; because both driving sides of Alpamayo 1.5 were measured together, its numbers do not satisfy the registration discipline, so the table is evaluated on the five policies that do. Retaining the replaced policy changes none of the seven verdicts.

a) What transferred: 4 of the seven predictions hold, and they are the orderings and verdicts. The night rendering outweighs the pedestrian in every policy. Exposure keeps its order ($\rho = 0.90$), and so does over-reaction, with SimLingo the most unstable. SimLingo collides most, with the visible and removed rates within 3 points of each other.

b) What did not: Three predictions fail.

- Point values drift: only 60% of the model–exam cells stay inside their Boston interval.
- In Singapore, DiffusionDrive and DiffusionDriveV2 show a small but non-zero hazard sensitivity (HS = 0.09 and 0.08).
- The “purity” hypothesis is reversed. We had predicted that policies whose lighting effect is more orthogonal to their pedestrian response would shift less between cities; the observed rank correlation is -0.90 . Alignment is itself unstable between the two cities (Section VIII).

c) Reading: A small check-up can therefore report an ordering and a verdict for use in a new domain, but not a calibrated value.

VII. Re-measuring the Axes in the New Domain

The transfer test of Section VI moved a diagnosis between cities. A more useful question for deployment is whether the method moves: can a handful of counterfactual cases in the new domain characterise how the policies behave there? We repeat both interventions on the Singapore near-pedestrian scenes, on the original images used by the benchmark, and read the two axes per scene: F , the plan displacement caused by removing the target pedestrian, and I , the displacement caused by the night rendering.

a) What the axes say per scene: Across the ten detailed cases and six policies (60 cells), the median F is 0.05 m and the median I is 0.62 m; the lighting moves the plan more in 85% of cells. Two rules follow directly

TABLE VII

Re-measuring the two axes in the new domain. CFR measured on Boston frames, on Singapore frames, and per scene on the ten Singapore cases (median). The verdict $\text{CFR} < 1$ holds on both sides for every policy, while the value drifts by 0.6–1.7 $\times$; the across-policy ordering of CFR agrees between sides only at $\rho = +0.37$ (specificity: $\rho = +0.83$). Right: share of near-pedestrian scenes in which each rule holds (P-F: a near-zero F axis implies removing the pedestrian leaves clearance and contact unchanged; P-I: $\text{CFR} < 1$ implies the night rendering moves the outcome at least as much as the removal). Both rules are two-sided: each cell gives the hit rate when the premise holds / when it is reversed (fwd/rev), so every scene receives a prediction, and the share of scenes whose minimum time-to-proximity to the pedestrian’s logged future falls below 1.5s. The large-sample columns use the 246 Singapore scenes all six policies share.

Policy	CFR			TTC	rule holds %	
	Bos.	Sing.	scene	<1.5s	P-F	P-I
DD	0.17	0.24	0.03	13%	95/74	95/97
LTF	0.34	0.19	0.12	16%	96/100	97/100
DDv2	0.39	0.52	0.13	22%	96/67	96/87
SimLingo	0.32	0.19	0.06	56%	93/73	95/91
AutoVLA	0.41	0.46	0.00	21%	95/60	98/86
Alpamayo 1.5	0.15	0.26	0.13	20%	99/33	94/93

from the axes. We state each in both directions, so that every cell receives a prediction rather than only those whose premise happens to hold:

- P-F: if $F < 0.5$ m, removing the pedestrian should leave the clearance and the contact outcome unchanged; if $F \geq 0.5$ m, it should change them.
- P-I: if $\text{CFR} < 1$, the night rendering should move the outcome at least as much as the removal; if $\text{CFR} \geq 1$, the removal should move it at least as much as the night.

The two directions do not fare equally. On the detailed cases, P-F holds in 54/55 cells in the forward direction but only 2/5 in the reverse; P-I holds in 51/51 and 8/8. Scaling both interventions to the 246 Singapore scenes that all six policies share reproduces the asymmetry exactly (Table VII): P-F holds in 93–99% of scenes forward against 33–100% reversed, while P-I holds in 94–98% and 86–100%. A near-zero F axis is therefore a reliable certificate that the pedestrian is not being used, but a large F is not a promise of safer behaviour.

b) When the plan does move, which way does it move?: The reverse direction fails for a specific and instructive reason. F measures how far a plan moves, not where it moves to, so we decompose it (Table VIII). Alongside F we read I , the displacement the same policy produces under an edit that has nothing to do with the pedestrian, which serves as its own jitter floor, and ΔS , the change in clearance to the pedestrian’s logged future. A response deserves to be called avoidance only if it satisfies all three conditions: the plan moved ($F \geq 0.5$ m), it moved more than the policy’s own jitter ($F > I$), and the movement bought distance ($\Delta S \geq 0.5$ m).

Three readings follow. First, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes, so most of

TABLE VIII

A plan that moves is not a plan that yields. For each policy and each of the 246 Singapore near-pedestrian scenes we measure F (displacement caused by removing the pedestrian), I (displacement caused by the night rendering, the policy’s own jitter under an irrelevant edit) and ΔS , the resulting change in clearance to the pedestrian’s logged future ($\Delta S > 0$: seeing the pedestrian keeps the plan farther away). $\eta = \Delta S/F$ is the share of the displacement that becomes separation, taken over the scenes in which the plan actually moved. Genuine avoidance requires all three: $F \geq 0.5$ m, $F > I$, and $\Delta S \geq 0.5$ m.

Policy	$F > I$	$F \geq 0.5$	median η	genuine	$\rho(\Delta S, a_{\text{req}})$
DD	12%	23	+1.92	12 (5%)	−0.05
LTF	4%	1	+2.70	1 (0%)	−0.37
DDv2	18%	27	+0.62	11 (4%)	−0.15
SimLingo	23%	62	−0.59	10 (4%)	−0.00
AutoVLA	11%	5	+0.00	1 (0%)	−0.01
Alpamayo 1.5	12%	3	+0.53	0 (0%)	+0.07

what F registers sits inside the policy’s own variability. Second, of the 1476 policy–scene cells, 35 (2.4%) are genuine avoidance by the definition above. Third, the displacement that does occur is not aimed: of the 121 cells in which the plan moved, 43 increased the clearance, 42 decreased it, and 36 left it unchanged. Per policy the split is sharply different. DiffusionDrive reacts in few scenes but in the right direction (14 away against 3 towards, median $\Delta S = +2.16$ m). SimLingo moves in more scenes than the other five combined and moves towards the pedestrian more often than away (32 against 13, median −0.59 m); it is also the policy with the highest rate of sub-1.5s time-to-proximity (Table VII). Finally, the clearance a policy yields does not grow with the hazard: the rank correlation between ΔS and the deceleration the scene demands is −0.37+0.07 across the six policies.

An exam that rewarded responsiveness alone would therefore rank the least safe policy first. This is the asymmetry we want from a check-up: it can clear a policy of using the pedestrian, and it refuses to credit a policy merely for reacting.

c) A small sample in the new domain ranks; it does not calibrate: We can now ask what a handful of new-domain cases actually buys. We draw k scenes at random from the shared pool, re-measure both axes on them, and compare the resulting estimate to the large-sample values, 200 draws per k . Two things behave very differently. The ordering of how dangerously the policies behave—the share of scenes whose minimum time-to-proximity falls below 1.5s—is recovered almost immediately: $\rho = +0.64$ at $k = 10$ and $\rho = +0.70$ at $k = 20$ (with $\rho \geq 0.6$ in 80% of draws), rising to $\rho = +0.76$ at $k = 40$. The level is not: the same $k = 20$ sample still misstates the violation rate by 12 points on average, and CFR by 0.116. The rule hit-rates sit in between; $k = 20$ estimates them to 3.6 points. A small check-up in a new domain therefore supports a ranking and a verdict, not a calibrated number—which is exactly the use we propose for it.

TABLE IX

What the night rendering does to the plan. CFR: plan change from removing the pedestrian over plan change from the night rendering, same frames (corridor pedestrians 5–12 m, 95% CI; ideal $\gg 1$). Speed: change of planned mean speed over 2.5 s under the night rendering. ΔS : change of whole-plan separation from the pedestrian (negative = closer). Both on frames where the ego moves (≥ 1 m/s). Align: direction of the night-induced change relative to the removal-induced one, Boston $\rightarrow$ Singapore (0 = orthogonal). Bold: $p < 0.01$.

Policy	CFR [95% CI]	speed	ΔS (m)	align
DD	0.30 [0.20,0.43]	+37%	−0.37	+0.55 $\rightarrow$ +0.09
LTF	0.36 [0.24,0.49]	+12%	−0.24	−0.09 $\rightarrow$ −0.42
DDv2	0.49 [0.33,0.70]	−8%	+0.15	+0.63 $\rightarrow$ +0.16
SimLingo	0.31 [0.21,0.44]	+16%	−0.47	+0.30 $\rightarrow$ +0.14
AutoVLA	0.38 [0.21,0.57]	0%	0	+0.65 $\rightarrow$ +0.62
Alpamayo 1.5	0.18 [0.10,0.30]	−2%	+0.03	+0.81 $\rightarrow$ −0.30

d) Re-measuring beats importing: Importing the Boston CFR values misstates the Singapore ones by 0.099 on average, about what re-measuring $k = 40$ fresh scenes achieves (0.078); doubling that to 80 scenes halves the error again (0.047 against 0.099 for the imported values).

e) Which exam travels: Of the exams in the report, the one that predicts the new domain’s behaviour is not the one that measures how much a policy reacts, but the one that measures how steadily it drives. Correlating each Boston exam against the Singapore rate of sub-1.5 s time-to-proximity over the six policies, specificity gives $\rho = -0.94$ —the only value significant at the two-sided 5% level for $n=6$ —against +0.77 for exposure and +0.37 for CFR. Specificity is the exam that asks how far a plan moves when an irrelevant part of the scene is edited; a policy that answers “not at all” is one whose trajectory is stable, and stability is what carries across cities. The ordering used here is the one measured on the counterfactual pool, whose readout excludes the scenes that carry no removal edit; on the larger original-image set of Section V exposure reaches 0.94 against this same target. Exposure therefore travels on both pools, but specificity is the stronger predictor and the only one that clears significance here. This is consistent with the decomposition above: displacement that is not aimed away from the pedestrian is not protection, and a policy that produces a lot of it is dangerous in both cities.

f) What transfers and what must be re-measured: The verdict $\text{CFR} < 1$ holds on both driving sides for every policy, but the value drifts by a factor of 0.6 to 1.7, and the across-policy ordering of CFR barely agrees between sides ($\rho = +0.37$, against +0.83 for specificity). A single scene is weaker still: the per-scene medians (Table VII) sit well below either aggregate. The practical reading is that the instrument transfers while its readings do not: entering a new domain, one re-measures the axes there with a few dozen scenes rather than importing numbers measured elsewhere.

VIII. Lighting: Magnitude, Direction, and the Pedestrian

Lighting should not change what a driver does, so this exam asks how far a policy is moved by an irrelevant change. We analyse it last. It is the most robust result in the check-up and also the one whose direction is least stable.

a) Magnitude: On the same frames, removing a pedestrian from the corridor moves every policy’s plan only 0.18–0.49 times as far as the night rendering does (Table IX). Every 95% interval lies below one. The result holds on detector-verified edits ($\text{CFR} = 0.30\text{--}0.64$), where only Alpamayo-1.5’s interval (45 frames) reaches above one. Tracing the rendering on a separate pool of 312 lead-vehicle events shows that the effect is carried by sensor noise rather than darkness. For DiffusionDrive, darkening alone raises the planned speed by +0.08 m/s and darkening with noise by +2.05 m/s. The effect is also global: perturbing only the sky, with the corridor untouched, recovers 19% of the effect of deleting the lead vehicle.

b) Direction: faster, and therefore closer: Darkness reduces visibility, so a policy that slows at night is doing something defensible; one that speeds up is not. On frames where the ego moves, three policies plan to go faster under the night rendering (Table IX): DiffusionDrive by 37%, SimLingo by 16% and LTF by 12%. The gain grows along the horizon; for DiffusionDrive it rises from +0.12 m/s in the first half-second to +1.03 m/s in the last. As a result, their whole plans pass 0.24–0.47 m closer to the pedestrian. The approach is longitudinal. The more the plan’s endpoint moves forward, the closer it gets (rank correlation -0.73 to -0.96), whereas the endpoint shifts toward the pedestrian’s side in no more than 47% of frames, which is no more often than chance. Contacts barely change, because most pedestrians remain beyond the 2.5 s plan. DiffusionDriveV2, the only policy that also reads LiDAR, is the one that slows at night (−8%) and keeps a larger distance. AutoVLA and Alpamayo-1.5 are unaffected. In representation space the night direction opposes the hazard direction (mean cosine -0.54 in four policies), so the policies encode “it got dark” as they encode “the hazard is gone”.

c) Orthogonality as a trait: We hypothesised that policies whose lighting effect is orthogonal to their pedestrian response would be more stable across domains. The data do not support this. Alignment changes between cities for most policies (DiffusionDrive +0.56 $\rightarrow$ +0.09, DiffusionDriveV2 +0.63 $\rightarrow$ +0.16). Only AutoVLA keeps it (+0.65 $\rightarrow$ +0.62). The purer policies shifted more, not less (Table VI, P6). The magnitude of the lighting effect is a trait that a small check-up can measure. Its direction is not, until hundreds of frames are available (Section VI).

IX. Discussion

a) Using the report: Read one exam at a time, the report answers separate questions that a single score

merges. Whether a policy is safe because it yields or because it stays short is answered by exposure together with hazard sensitivity: DiffusionDrive keeps its distance by planning short trajectories, which holds only as long as the trajectories stay short, whereas AutoVLA and Alpamayo-1.5 cover a human-like distance without the response to match. Whether a plan that moves is a plan that yields is answered by the decomposition of Section VII: across 246 scenes and six policies, only 2.4% of cells meet all three conditions, and displacement alone would have ranked the least safe policy first. The exam that no current benchmark reports is scaling: whether the response grows when the hazard does. Five policies fail it, and the one that passes, DiffusionDriveV2, does so at a sensitivity below 0.15.

b) Limitations: The exams are open-loop. The approach sequences are replayed from logs and re-anchored to the logged pose. The night is a synthetic rendering that also halves the number of other pedestrians the detector finds. The counterfactual pairs are constructed by editing, not captured, and so carry noise: 8.5% of removals leave a detectable residue, the inpainting artefact is 5–17% of the effect, and the VLA models can be edited only through their front camera. Alpamayo-1.5 covers 170 of the 236 frames. SimLingo plans 2.5s; for NAVSIM’s 4s we extrapolate at constant velocity (holding position instead gives EPDMS 0.47, still the lowest). Cross-policy correlations rest on six policies.

c) Future work: The check-up is designed as the first half of a benchmark-to-deployment pairing. The next step is to collect paired data rather than construct it: the same scene with and without a pedestrian, by day and by night. We then plan to validate the diagnoses against the same policies’ behaviour on a real vehicle. The sample-size laws above tell us how large that collection must be. A few dozen pairs suffice for planned distance, over-reaction and lighting; hundreds of pairs with a genuine need to react are needed for hazard sensitivity.

X. Conclusion

A benchmark score reports an outcome, not a cause. We read the behaviour that a deployment decision needs from a few hundred logged frames, using counterfactual exams that an independent detector verifies. Three results are worth carrying forward. A near-zero counterfactual response is a reliable certificate that the pedestrian did not enter the decision, while a large response certifies nothing: of the plans that moved, as many moved towards the pedestrian as away from it. What transfers between cities is how steadily a policy drives rather than how strongly it reacts, and that exam is the only one—among ours and the standard scores alike—that predicts the new city’s behaviour at significance. And each verdict can be priced in advance: some are settled in a dozen frames, others need hundreds, which is what a small check-up must know about itself before it is used to decide anything.

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